

Curriculum Vitae

Shahabuddin Khan, M.Sc.

PhD Student and Graduate Research Assistant,
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Former Head of the Department,
Department of Electrical Engineering
Western Region Campus, IOE, TU



Objective:

To select a career which, in tune with my aptitude and ability, a job that ensures me of dignity and motivation, ambience and upward mobility. As an engineer my endeavor would be to strive for perfection in any assignment I take.

Academic Qualification:

S.NO	DEGREE	COLLEGE/UNIVERSITY	YEAR OF COMPLETION	PERCENTAGE
1.	M.Sc. (Distributed Generation)	Pashchimanchal Campus/ T.U.	2018	85.67%
2.	B.E.(Electrical Engineering)	Pashchimanchal Campus/ T.U.	2015	73.39%
3.	High School(12 th)	New Horizon H. S. School/HSEB	2011	73.10%
4.	SLC(10 th)	Zalmonah Little Angel English Boarding School	2008	80.13%

Academic Experience:

Graduate Research Assistant

Aug. 2024- till now

- Working as Graduate Research Assistant at “Rensselaer Polytechnic Institute” under the supervision of Prof. Zheyu Zhang for the thermal and loss modelling of IGBT for reliability assessment.

Lecturer, Pulchowk Campus

March 2021-Aug. 2024

- Working as Lecturer in “Pulchowk Campus” supervising students and research on Distributed Generation and Control System.

- Working as Lecturer in “Pulchowk Campus” conducting classes of Reliability Engineering of Power Electronics and Drives, Transmission and Distribution, Reliability Engineering, Control System and Basic Electrical Engineering.

Lecturer, Pashchimanchal Campus

April, 2018-March, 2021

- Worked as Lecturer in “Pashchimanchal Campus” teaching subjects like Power System Reliability, Advance Power Electronics Energy Audit and Applied Photovoltaic System.
- Effectively Supervising Research Graduate in Distributed Generation.

Assistant Lecturer, Pashchimanchal Campus

Nov. 2017-April, 2018

- Worked as Assistant Lecturer in “Pashchimanchal Campus” teaching subjects like Power System Reliability, Utilization of Electrical Energy and Electrical Machine Design.

MATLAB Trainer, Pashchimanchal Campus

- Worked as MATLAB trainer in “Pashchimanchal Campus”.
- Proficient in using computer and the internet for educating.
- Providing MATLAB training material and programs.

Research Profile:

Thesis: Co-ordination of directional over-current relays in distribution network with DG considering transient stability constraint

This thesis is aimed to determine the optimal relay coordination of overcurrent relay considering transient stability as a constraint to avoid the effect of DG on protection system.

Tool: MATLAB, DigSilent

Project: Four Quadrant operation of DC motor interface with PC

This project is aimed to develop a control mechanism to control the speed of DC motor along with performing four quadrant operation of DC motor. The motor is controlled by a PC.

Tool: Arduino

Conferences:

1. **Session Chairperson** on “International Conference on Role of Energy for Sustainable Social Development in the New Normal Era”, organized by IEEE PES Nepal Chapter on 28-29 Dec 2020, Kathmandu
2. **Session Chairperson** on IOE Graduate Conference, Dec 13-14, 2019 at Pashchimanchal Campus Pokhara

3. **Participant** on “4th International Conference on Renewable Energy Technology for Rural and Urban Development (RETRUD-18)” Renewable Energy Forward for Economic and Sustainable Development 12 Kartik- 14 Kartik, Kathmandu
4. **Participant** on 10th International Conference on Quality, Reliability, INFOCOM Technology and Business Operations (Towards Smart, Reliable and Sustainable Future) 3rd Asadh-6th Ashadh, Kathmandu

Publications:

1. B. Kafle and **S. Khan**, "Comparison between PSO and Fuzzy PID Controller for Speed Control of DC Motor.", *Udbodhan*, TU IOE Pashchimanchal Campus, Vol 5, Issue 1, 5 October 2021.
2. **Khan, Shahabuddin**, and Ananta Adhikari. "Coordination of Over-Current Relays in Power Distribution Networks with DG Considering Transient Stability Constraint." *Proceedings of IOE Graduate Conference*, IOE, TU, Nepal, Vol 7, 2019-Winter, 79-83.
3. Adhikari, Utsav, **S. Khan**, et al. "Battery Energy Storage System Optimization for Grid-Connected Wind-PV Hybrid System." *Proceedings of IOE Graduate Conference*, IOE, TU, Nepal, Vol 7, 2019-Winter, 21-27.
4. Giri, P., Dhakal, **S. Khan**, et.al. “Transformer Design using MATLAB” in *Proceedings of IOE Graduate Conference*, IOE, TU, Nepal, Vol 7, 2019-Winter, 53-58.
5. H. Karn, P. K. Pankaj, M. Karki, **S. Khan**. “Distributed Generation Allocation considering Uncertainties”, *Proceedings of IOE Graduate Conference*, IOE, TU, Nepal, Vol 8, 2020-Summer, 1-8.
6. **S. Khan**, M. Karki, “LPS Design Analysis: A Case Study of RIC Building at Pashchimanchal Campus”, *International Conference on Role of Energy for Sustainable Social Development in ‘New Normal’ Era*, 28th-29th December 2020, Kathmandu, Nepal
7. B. Sharma, M. Karki, **S. Khan**, B. Pokhrel. “DG Penetration Analysis on the Locational Marginal Pricing: A Case Study of INPS”, *2020 IEEE International Women (WIE) in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE)*, 26th-27th December, 2020, Bhubaneswar, India.
8. S. Bhattarai, **S. Khan**, B. Kafle, “Comparative Control of Doubly-Fed Induction Generator Using Vector Control and Model Predictive Control”, *Udbodhan*, TU IOE Pashchimanchal Campus, Vol 6, January 2021.
9. R. Gyawali, **S. Khan**, M. Karki, S. Regmi, “Impact analysis of Electric Vehicle with V2G and G2V service in Distribution Network”, *Proceedings of IOE Graduate Conference*, IOE, TU, Nepal, Vol 9, 2021.
10. Mandal, Tushar, and **Shahabuddin Khan**. "Anti-Islanding Protection in Power Distribution Network Consisting of Multi-DG Units and PHEVs Charging Stations using WPT and PNN." *IOE Grad. Conf.* Vol. 8914. 2023.
11. Phuyal, S., Shrestha, S., Sharma, S., Subedi, R., & **Khan, S.** (2023, March). Predictive Load Management Using IoT and Data Analytics. In *International Conference on Artificial Intelligence of Things* (pp. 153-168). Cham: Springer Nature Switzerland.
12. Sapkota, D. B., Neupane, P., Joshi, M., & **Khan, S.** (2024). Deep learning model for enhanced power loss prediction in the frequency domain for magnetic materials. *IET Power Electronics*.

Presentations:

1. A Resource Person on Webinar on Lightning Protection: Necessity in Nepal by Electrical Engineering Association, May 30, 2020.
2. A Resource Person on Webinar on Working with templates for making Papers and Research Articles by LaTeX by Department of Geomatics Engineering, M.Sc. Program, August 5, 2021.

Projects:

- **Young Scientist Grant (2023-2024):** Awarded by the Nepal Academy of Science and Technology for innovative research on the impact of e-cookers on energy distribution systems.

Personal Dossier:

Date of Birth	:	20-12-1993
Father's Name	:	Abdul Kalam
Mother's Name	:	Siratun Nisha
Temporary Address	:	55 Brunswick Ave, Troy, 12180
Permanent Address	:	Siswa-03, Kapilvastu, Nepal
Languages Known	:	English, Nepali, Awadhi and Hindi